

↑ DIMENSION MISMATCH ↓

DGP COMPLEXITY →

Independent

Correlated

Functionally
constrained

Statistical validity

(D1)

(D2)

(D3 / D4)

(m/n ratio)

Overcomplete
 $m > d$
(E5–E8)

FN: distributed codes

MCC-S
MCC-P, DCI-D, R²

FN: correlation +
distributed codes

MCC-P, MCC-S, DCI-D, R²

FN: redundancy +
distribution compound

All coordinate-wise

$m/n \geq 1$

Most metrics report
noise as signal

*Sparse autoencoders
operate here*

Matched
 $m = d$
(E1–E3)

Metrics calibrated

MCC-P, MCC-S, R², DCI-D
+ T-MEX, InfoM

FN: correlation
confound

MCC-P, MCC-S, R²
DCI-D
+ T-MEX

FN: predictability
conflation

MCC-P, MCC-S
DCI-D, R²

$0.1 \leq m/n < 1$

MCC-P, MCC-S inflate

DCI-D mild FP

Undercomplete
 $m < d$
(E4)

FN: partial recovery
penalised

MCC-P, MCC-S, R²
DCI-D
+ T-MEX

FN: partial recovery +
correlation confound

MCC-S
MCC-P, DCI-D, R²

FN: lossless vs lossy
indistinguishable

All metrics

$m/n < 0.1$

Metrics calibrated

Low FP risk

m/n ↑



Calibrated



Some metrics fail



Most metrics fail



All metrics fail

R², T-MEX: robust to false-positive inflation